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Techno-Economic Assessment of On-Site Hydrogen Refueling Stations in Italy

Levelized Cost of Hydrogen (LCOH) Analysis

Simplified Case Study based on Minutillo et al. (2021)

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Abstract

This report presents a simplified techno-economic analysis of on-site Hydrogen Refueling Stations (HRS) that produce hydrogen locally through alkaline water electrolysis, powered by a combination of grid electricity and a grid-connected photovoltaic (PV) plant. The study reproduces and adapts the methodology of Minutillo et al. (2021) for the Italian scenario, applying four simplifications prescribed by the course instructor: (i) revenue from selling excess PV electricity is excluded; (ii) water purchase cost is neglected; (iii) hydrogen storage tank cost is excluded; and (iv) the total plant electricity demand is split with a fixed 90/10 rule between electrolysis and the compression-plus-cooling load. The Levelized Cost of Hydrogen (LCOH) is computed for twelve configurations — three plant sizes (Micro 50, Small 100, Medium 200 kg/day) and four electricity mixes (100%, 75%, 50% and 25% grid) — at two Italian sites, Naples (South) and Milan (North). The results range from **6.50 €/kg** (Medium, 25% grid, Naples) to **11.73 €/kg** (Micro, 100% grid), confirming strong economies of scale and the benefit of higher renewable penetration. Because PV export revenue is removed, the optimum shifts to the lowest grid share, unlike the original paper where a 50% grid mix was optimal. A sensitivity analysis identifies electrolyzer efficiency and grid electricity price as the dominant technical and economic levers, and a carbon-intensity estimate shows the emissions fall from 15.7 to 5.6 kg CO₂/kg H₂ as the PV share increases.

Keywords: Hydrogen Refueling Station, LCOH, alkaline electrolysis, PV integration, techno-economic analysis, Italian scenario, sensitivity analysis.

1. Introduction

The deployment of hydrogen fuel-cell electric vehicles (FCEVs) depends critically on the availability of refueling infrastructure. Among the possible configurations, on-site hydrogen production — where hydrogen is generated at the station itself through water electrolysis rather than produced elsewhere and transported — is attractive because it removes delivery logistics and allows direct integration with local renewable generation. The economic viability of such stations is captured by a single indicator, the Levelized Cost of Hydrogen (LCOH), expressed in €/kg H₂, which consolidates the capital investment, the operation and maintenance expenditure, and the mid-life component replacement into one comparable figure.

This work is a student-level replication of the techno-economic assessment carried out by Minutillo et al. (2021) for the Italian context. Following the instructor's brief, four explicit simplifications are applied to obtain a transparent and reproducible model while preserving the analytical logic of the original study:

Rule	Simplification	Effect on the model
S1	No revenue from PV export	The revenue term is set to zero in the LCOH formula (paper: 50 €/MWh).
S2	No water purchase cost	Water OPEX neglected as negligible against electricity cost.
S3	No hydrogen storage CAPEX	Storage tank cost excluded (paper: 1,644 €/kg H ₂).
S4	90/10 energy rule	Total demand = electrolyzer demand / 0.90; the remaining 10% covers compression and cooling.

Table 1 — The four simplifications applied relative to Minutillo et al. (2021).

The objectives are: (a) to model the energy demand under the 90/10 rule; (b) to compute CAPEX, OPEX and replacement cost for all twelve configurations; (c) to calculate the LCOH and identify the optimal configuration; (d) to compare a southern and a northern Italian site; and (e) to perform a sensitivity analysis and a first-order carbon-intensity estimate.

2. System Description and Plant Configurations

The HRS consists of four subsystems connected in series. The **electrolysis unit** (alkaline technology) produces hydrogen from demineralised water at 10 bar. The **compression unit** raises the pressure from

10 bar to 820 bar for cascade storage filling. The **refrigeration unit** pre-cools the hydrogen to $-40\text{ }^{\circ}\text{C}$, complying with the SAE J2601 fueling protocol (which limits the in-tank temperature to $85\text{ }^{\circ}\text{C}$). Finally, the **dispenser** delivers hydrogen to the vehicle at 700 bar. Electricity is supplied jointly by the national grid and an on-site grid-connected PV plant.

The electrolyzer is modular; each base module produces 50 kg/day. The Small and Medium plants use two and four modules respectively. The rated module parameters are summarised in Table 2.

Parameter	Value	Unit
H ₂ production per module	50	kg/day
H ₂ flow rate	23	Nm ³ /h
Nominal AC power	118	kW
Specific energy consumption (baseline)	5.1	kWh/Nm ³
H ₂ density	0.089	kg/Nm ³
LHV of hydrogen	33.0	kWh/kg
Electrolyzer efficiency η	57.6	%
Operating pressure	10	bar
Electrolyzer cost	1,100	€/kW

Table 2 — Alkaline electrolyzer module parameters (Minutillo et al., 2021).

The electrolyzer efficiency is defined as the ratio of the hydrogen lower heating value to the specific electrical energy input per unit mass: $\eta = \text{LHV} / (w_{\text{spec}} / \rho) = 33.0 / (5.1 / 0.089) = 57.6\%$.

Three plant sizes and four electricity-supply strategies define the twelve configurations analysed (Table 3). The electricity mix is expressed as the annual share of energy drawn from the grid, the remainder being supplied by PV.

Plant size	Full Grid (100%)	High Grid (75%)	Mid Grid (50%)	Low Grid (25%)
Micro — 50 kg/day	Micro_FG	Micro_HG	Micro_MG	Micro_LG
Small — 100 kg/day	Small_FG	Small_HG	Small_MG	Small_LG
Medium — 200 kg/day	Medium_FG	Medium_HG	Medium_MG	Medium_LG ★

Table 3 — The twelve plant configurations (3 sizes $\times$ 4 electricity mixes). ★ = lowest LCOH in the simplified model.

3. Methodology

3.1 Energy demand (90/10 rule)

The annual electricity consumption of the electrolyzer is obtained from its rated power and the yearly operating hours (continuous 24 h operation, 8,760 h/year): $E_{elec} = N_{modules} \times 118 \text{ kW} \times 8,760 \text{ h} \div 1,000$. Under Simplification S4, the total plant demand is $E_{total} = E_{elec} / 0.90$, so that compression and cooling jointly account for the remaining 10%. The compressor power itself is obtained from the isentropic specific work between 10 and 820 bar, giving 8.3, 16.7 and 33.3 kW for the three sizes; the refrigerator is sized at 16.3 kW per dispenser. Table 4 reports the resulting energy balance and the required PV capacity for the Naples site.

3.2 PV plant sizing

The PV plant is sized to deliver the required annual energy share: $P_{PV} [\text{kWp}] = E_{PV} [\text{MWh/yr}] \times 1,000 / E_{specific} [\text{kWh/kWp/yr}]$. Two sites are compared using annual specific yields obtained directly from PVGIS (SARAH3 database, crystalline silicon, optimised tilt and azimuth, 14% system loss): Naples (South, 1,497 kWh/kWp/yr) and Milan (North, 1,369 kWh/kWp/yr). The lower northern yield requires a larger PV plant for the same energy contribution. Both sites were evaluated with identical PVGIS settings for consistency; the original paper reported 1,272 kWh/kWp/yr for Naples using the SAM tool, which explains the small difference in the absolute figures.

3.3 LCOH formula

Incorporating simplifications S1, S2 and S3, the LCOH reduces to the annualised total cost divided by the annual hydrogen output:

$$LCOH = (C_{inv,a} + C_{rep,a} + C_{O\&M}) / M_{H_2} \quad [€/kg \text{ H}_2]$$

The annualised capital cost is $C_{inv,a} = CRF \times C_{inv}$, where the Capital Recovery Factor for $i = 3\%$ and $n = 20$ years is $CRF = i(1+i)^n / [(1+i)^n - 1] = 0.0672$. The annualised replacement cost discounts each component replacement to the present: the electrolyzer (at 40.5% of its CAPEX), compressor, dispenser and water auxiliary are replaced in year 10, and the refrigerator in year 15; storage replacement is excluded (S3). The O&M term includes component maintenance and the annual grid-electricity purchase. The annual hydrogen output is $M_{H_2} = \text{capacity} [\text{kg/day}] \times 365$.

Configuration	E_{elec} (MWh/yr)	E_{total} (MWh/yr)	Grid frac.	E_{grid} (MWh/yr)	E_{PV} (MWh/yr)	PV size (kWp)
Micro_FG	1,033.7	1,148.5	100%	1,148.5	0	0
Micro_HG	1,033.7	1,148.5	75%	861.4	287.1	192
Micro_MG	1,033.7	1,148.5	50%	574.3	574.3	384
Micro_LG	1,033.7	1,148.5	25%	287.1	861.4	575
Small_FG	2,067.4	2,297.1	100%	2,297.1	0	0
Small_HG	2,067.4	2,297.1	75%	1,722.8	574.3	384
Small_MG	2,067.4	2,297.1	50%	1,148.5	1,148.5	767
Small_LG	2,067.4	2,297.1	25%	574.3	1,722.8	1,151
Medium_FG	4,134.7	4,594.1	100%	4,594.1	0	0
Medium_HG	4,134.7	4,594.1	75%	3,445.6	1,148.5	767
Medium_MG	4,134.7	4,594.1	50%	2,297.1	2,297.1	1,534
Medium_LG	4,134.7	4,594.1	25%	1,148.5	3,445.6	2,302

Table 4 — Energy demand and PV sizing for all configurations (Naples, 90/10 rule).

4. Cost Data and Economic Assumptions

All cost parameters are taken from Minutillo et al. (2021). Items removed by the simplifications (storage, water purchase, PV revenue) are shown for transparency. The grid electricity price depends on the annual electricity consumption tier.

Parameter	Value	Unit	Used?
Plant lifetime	20	years	Yes
Nominal interest rate	3	%	Yes
Electrolyzer cost	1,100	€/kW	Yes
PV plant cost	950	€/kWp	Yes
Compressor cost	43,872·P ^{0.5861}	€	Yes
Refrigerator cost	5,374	€/kW	Yes
Dispenser cost	65,000	€/unit	Yes
Water auxiliary cost	8.47	€/kW	Yes
Storage system cost	1,644	€/kg H ₂	No (S3)
O&M electrolyzer / compressor	2 / 8	% CAPEX/yr	Yes
O&M refrigerator / dispenser	3 / 3	% CAPEX/yr	Yes
O&M PV plant	1.58	% CAPEX/yr	Yes
Electricity price (<1 / <2.5 / <5 GWh)	129 / 119 / 109	€/MWh	Yes
PV excess sell price	50	€/MWh	No (S1)
Water purchase	1.006	€/m ³	No (S2)
Replacement year (elec./comp./disp.)	10	year	Yes
Replacement year (refrigerator)	15	year	Yes

Table 5 — Economic parameters. Grey rows are excluded by the indicated simplification.

5. Financial Analysis

5.1 Capital expenditure (CAPEX)

CAPEX comprises the electrolyzer, compressor, refrigerator, dispenser(s), water auxiliary system and PV plant; storage is excluded (S3). It grows with plant size and with PV share, since a larger PV plant requires proportionally higher investment (Table 6).

Config.	Electrolyzer	Compressor	Refrig.	Disp.	Water	PV plant	TOTAL
Micro_FG	129.8	152.0	87.7	65.0	1.0	0.0	435.5
Micro_HG	129.8	152.0	87.7	65.0	1.0	182.2	617.7
Micro_MG	129.8	152.0	87.7	65.0	1.0	364.4	799.9
Micro_LG	129.8	152.0	87.7	65.0	1.0	546.6	982.1
Small_FG	259.6	228.2	175.3	130.0	2.0	0.0	795.1
Small_HG	259.6	228.2	175.3	130.0	2.0	364.4	1,159.6
Small_MG	259.6	228.2	175.3	130.0	2.0	728.9	1,524.0
Small_LG	259.6	228.2	175.3	130.0	2.0	1,093.3	1,888.4
Medium_FG	519.2	342.6	175.3	130.0	4.0	0.0	1,171.1
Medium_HG	519.2	342.6	175.3	130.0	4.0	728.9	1,900.0
Medium_MG	519.2	342.6	175.3	130.0	4.0	1,457.7	2,628.8
Medium_LG	519.2	342.6	175.3	130.0	4.0	2,186.6	3,357.7

Table 6 — CAPEX breakdown for all configurations (k€, Naples). Storage excluded (S3).

5.2 Operating expenditure (OPEX)

Annual OPEX consists of component O&M and the cost of purchased grid electricity; water cost is excluded (S2). Grid electricity is by far the dominant item and decreases as the PV share rises. The applicable price tier follows the annual electricity consumption (129 €/MWh for Micro, 119 for Small, 109 for Medium).

Config.	Electricity purchase	O&M (all components)	TOTAL OPEX
Micro_FG	148.2	19.4	167.5
Micro_HG	111.1	22.2	133.4
Micro_MG	74.1	25.1	99.2
Micro_LG	37.0	28.0	65.0
Small_FG	273.4	32.6	306.0
Small_HG	205.0	38.4	243.4
Small_MG	136.7	44.2	180.8
Small_LG	68.3	49.9	118.3
Medium_FG	500.8	47.0	547.8
Medium_HG	375.6	58.5	434.1
Medium_MG	250.4	70.1	320.4
Medium_LG	125.2	81.6	206.8

Table 7 — Annual OPEX for all configurations (k€/yr). Water cost = 0 (S2), Revenue = 0 (S1).

5.3 Replacement cost (REPLEX)

Replacement covers the electrolyzer (40.5% of CAPEX, year 10), the compressor, dispenser and water auxiliary (year 10) and the refrigerator (year 15). No storage replacement is included. The undiscounted REPLEX depends only on plant size (Table 8); its annualised, discounted value enters the LCOH.

Plant size	Electrolyzer	Comp. + Disp. + Water	Refrigerator	TOTAL (undiscounted)	C_rep,a (annualised)
Micro (50 kg/d)	52.6	218.0	87.7	358.3	17.3
Small (100 kg/d)	105.1	360.2	175.3	640.6	30.8
Medium (200 kg/d)	210.3	476.6	175.3	862.2	41.9

Table 8 — Replacement cost by plant size (k€). Same across electricity mixes.

6. LCOH Results and Discussion

6.1 LCOH — Naples (South Italy)

Table 9 and Figure 1 report the LCOH for all twelve configurations at Naples. The global minimum is **6.50 €/kg** (Medium_LG) and the maximum is **11.73 €/kg** (Micro_FG).

Plant size	Full Grid 100%	High Grid 75%	Mid Grid 50%	Low Grid 25%
Micro (50 kg/d)	11.73	10.53	9.33	8.13
Small (100 kg/d)	10.69	9.65	8.61	7.56
Medium (200 kg/d)	9.16	8.27	7.38	6.50

Table 9 — LCOH (€/kg H₂), simplified model, Naples. Green = minimum, red = maximum.

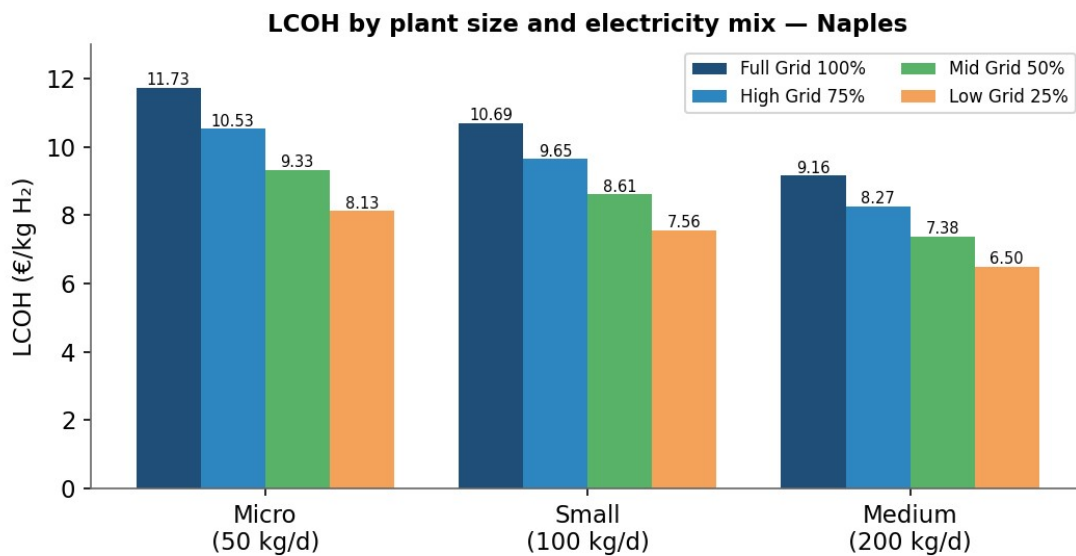


Figure 1 — LCOH by plant size and electricity mix (Naples).

Three trends emerge. First, **economies of scale**: the Medium plant is 17–21% cheaper than the Micro plant for the same mix. Second, the **grid-fraction effect**: because PV export revenue is removed (\$1), the LCOH decreases monotonically as the grid share falls — every kWh shifted from grid to PV lowers OPEX with no offsetting revenue loss, so the Low Grid (25%) mix is optimal for every size. This differs from Minutillo et al. (2021), where the 50% mix was optimal precisely because the surplus PV was sold. Third, **OPEX dominance**: grid electricity is the largest cost driver, especially in the full-grid cases.

6.2 Annualised cost breakdown (best configuration)

For the best configuration, Medium_LG, the total annualised cost is 474.4 k€/yr against 73,000 kg/yr of hydrogen, giving 6.50 €/kg. The breakdown (Table 10, Figure 2) shows that annualised CAPEX now exceeds the grid-electricity bill, reflecting the large PV plant in a PV-heavy configuration.

Component	k€/yr	% of total
Annualised CAPEX	225.7	47.6%
Grid electricity	125.2	26.4%
O&M (all components incl. PV)	81.6	17.2%
Annualised REPLEX	41.9	8.8%
TOTAL	474.4	100%

Table 10 — Annualised cost breakdown, Medium_LG, Naples.

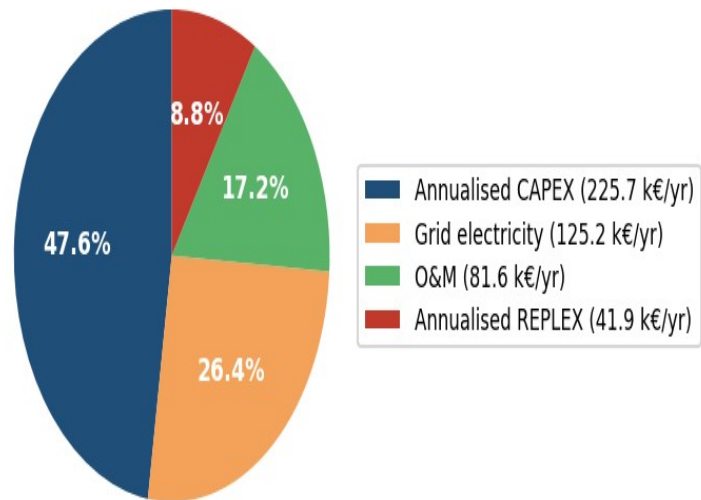
Annualised cost breakdown — Medium_LG (best)

Figure 2 — Annualised cost breakdown for the best configuration.

6.3 Location comparison: Naples vs. Milan

For PV-integrated configurations, the weaker solar resource in Milan (1,369 vs 1,497 kWh/kWp/yr) requires a larger PV plant to deliver the same annual energy, raising CAPEX and therefore LCOH. Full-Grid configurations (no PV) are identical at both sites. Because the two yields differ by only about 9%, the penalty is modest — around +2% to +4% for the PV-heavy Low Grid cases (Table 11, Figure 3).

Plant size	Naples FG	Milan FG	Naples HG	Milan HG	Naples MG	Milan MG	Naples LG	Milan LG
Micro	11.73	11.73	10.53	10.61	9.33	9.49	8.13	8.36
Small	10.69	10.69	9.65	9.73	8.61	8.76	7.56	7.79
Medium	9.16	9.16	8.27	8.35	7.38	7.54	6.50	6.73

Table 11 — LCOH comparison Naples vs. Milan (€/kg H₂). Full-Grid values coincide (no PV).

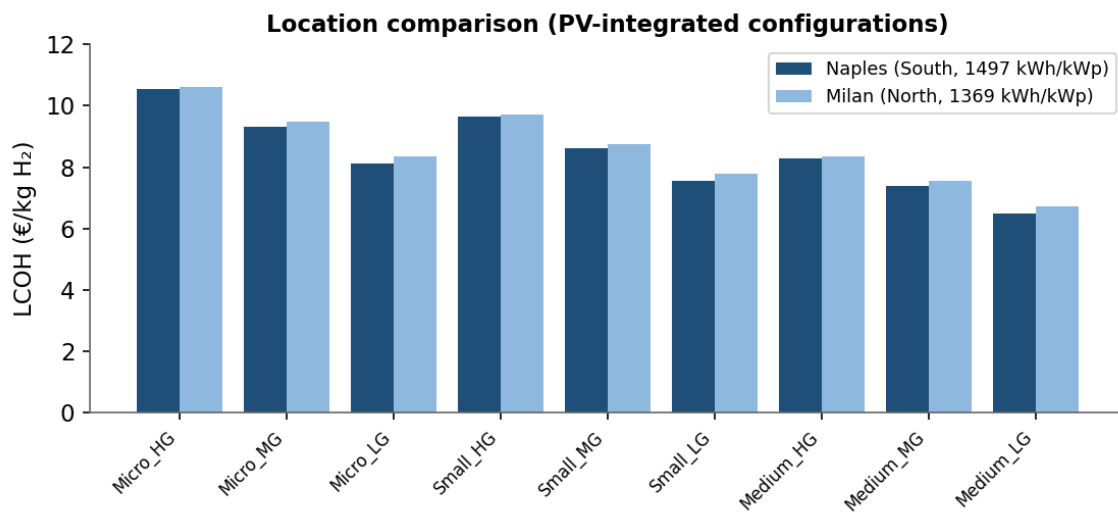


Figure 3 — Location comparison for PV-integrated configurations.

7. Sensitivity Analysis

The sensitivity analysis uses Medium_MG as the reference configuration. Two parameters are varied: the electrolyzer specific consumption (a technical lever) and the grid electricity price (an economic lever).

7.1 Electrolyzer specific energy consumption

The baseline alkaline electrolyzer consumes 5.1 kWh/Nm³ ($\eta = 57.6\%$). Modern PEM systems reach 4.5–4.8 kWh/Nm³ and advanced units approach 4.0 kWh/Nm³. Reducing consumption lowers both the electrolyzer power (hence CAPEX) and the electricity demand, so the LCOH falls almost linearly (Table 12).

Spec. consumption (kWh/Nm ³)	Efficiency	LCOH (€/kg)	Change vs. baseline	Technology
4.0	73.4%	6.09	−17.1%	Advanced PEM
4.5	65.3%	6.66	−9.4%	Commercial PEM
4.8	61.2%	7.01	−4.6%	Efficient PEM
5.1 (baseline)	57.6%	7.35	—	Alkaline
5.5	53.4%	7.81	+6.3%	Low-eff. alkaline
6.0	48.9%	8.38	+14.0%	Obsolete

Table 12 — LCOH sensitivity to electrolyzer specific consumption (Medium_MG, Naples).

7.2 Grid electricity price

The grid electricity price is the dominant OPEX driver. Around the 109 €/MWh baseline, each 10 €/MWh shifts the LCOH by roughly 0.3 €/kg. A Power Purchase Agreement at 70 €/MWh brings the LCOH down to about 6.16 €/kg, approaching the cost range of grey hydrogen from steam methane reforming (Table 13).

Electricity price (€/MWh)	LCOH (€/kg)	Change vs. baseline	Interpretation
70	6.16	-16.5%	PPA / renewables
90	6.79	-8.0%	Favourable contract
109 (baseline)	7.38	—	Italian tariff (>2.5 GWh)
130	8.05	+9.1%	Standard tariff
150	8.67	+17.5%	Energy-crisis level
180	9.62	+30.4%	Peak-price scenario

Table 13 — LCOH sensitivity to grid electricity price (Medium_MG, Naples).

Sensitivity analysis — reference Medium_MG

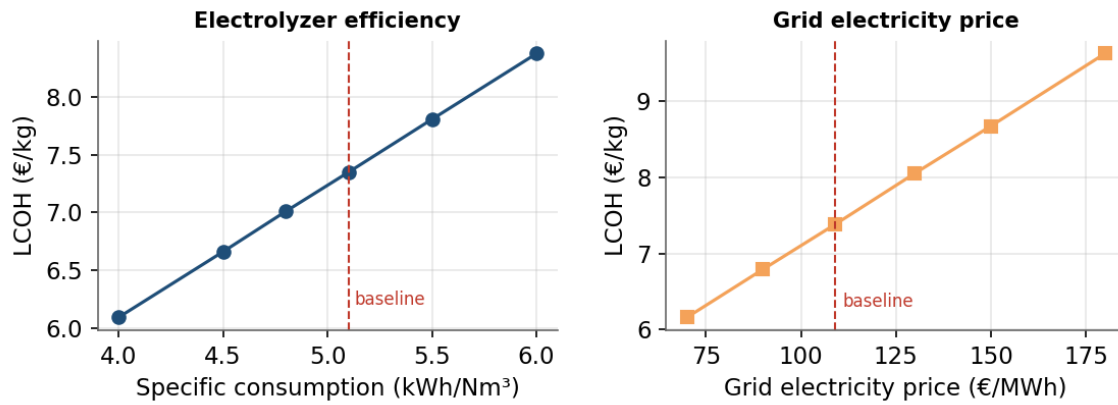


Figure 4 — Sensitivity of LCOH to electrolyzer efficiency (left) and grid electricity price (right).

8. Carbon Intensity (complementary analysis)

As a complementary indicator, the carbon intensity of the produced hydrogen is estimated by weighting the grid and PV energy shares by their emission factors — 250 gCO₂/kWh for the Italian grid and 35 gCO₂/kWh for PV on a life-cycle basis. The result depends only on the electricity mix, not on plant size, and falls from 15.7 kg CO₂/kg H₂ in the full-grid case to 5.6 kg CO₂/kg H₂ at 75% PV (Table 14, Figure 5). The Low Grid mix therefore minimises both cost and emissions, reinforcing its selection as the preferred configuration.

Electricity mix	Grid share	PV share	Carbon intensity (kg CO ₂ /kg H ₂)
Full Grid	100%	0%	15.7
High Grid	75%	25%	12.4
Mid Grid	50%	50%	9.0
Low Grid	25%	75%	5.6

Table 14 — First-order carbon intensity by electricity mix.

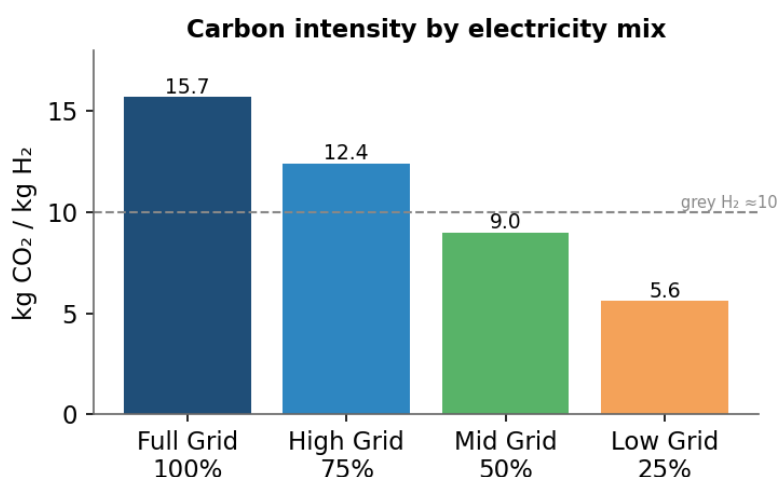


Figure 5 — Carbon intensity by electricity mix.

9. Conclusions

This simplified techno-economic assessment of on-site hydrogen refueling stations in Italy leads to the following conclusions:

- **Optimal configuration:** the Medium plant (200 kg/day) with Low Grid supply (25% grid, 75% PV) at Naples achieves the lowest LCOH, 6.50 €/kg H₂.
- **Economies of scale:** scaling from Micro to Medium reduces the LCOH by about 17–21% for any given electricity mix.
- **Role of PV without revenue:** once PV export revenue is removed (S1), a higher PV share always reduces LCOH, so the optimum shifts to the minimum grid share — in contrast to the original paper, where a 50% grid mix was optimal.
- **Dominant cost:** grid electricity is the single largest cost item, confirming that electricity pricing is the most critical economic parameter.
- **Location:** the southern site (Naples) outperforms the northern site (Milan) by roughly 2–4% on PV-heavy configurations, reflecting the modest ~9% difference in PVGIS annual yield between the two sites.

- **Technology and price levers:** switching to commercial PEM electrolysis (4.5 kWh/Nm³) lowers LCOH to 6.66 €/kg, and a 70 €/MWh PPA lowers it to 6.16 €/kg; combined, they push the cost below 6 €/kg.
- **Emissions:** the carbon intensity falls from 15.7 to 5.6 kg CO₂/kg H₂ as the PV share increases, so the least-cost configuration is also the least-emitting.

Future reductions in electrolyzer cost (projected below 400 €/kW by 2030), declining PV cost and grid decarbonisation are expected to lower the LCOH further, accelerating the transition to cost-competitive green-hydrogen mobility infrastructure.

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